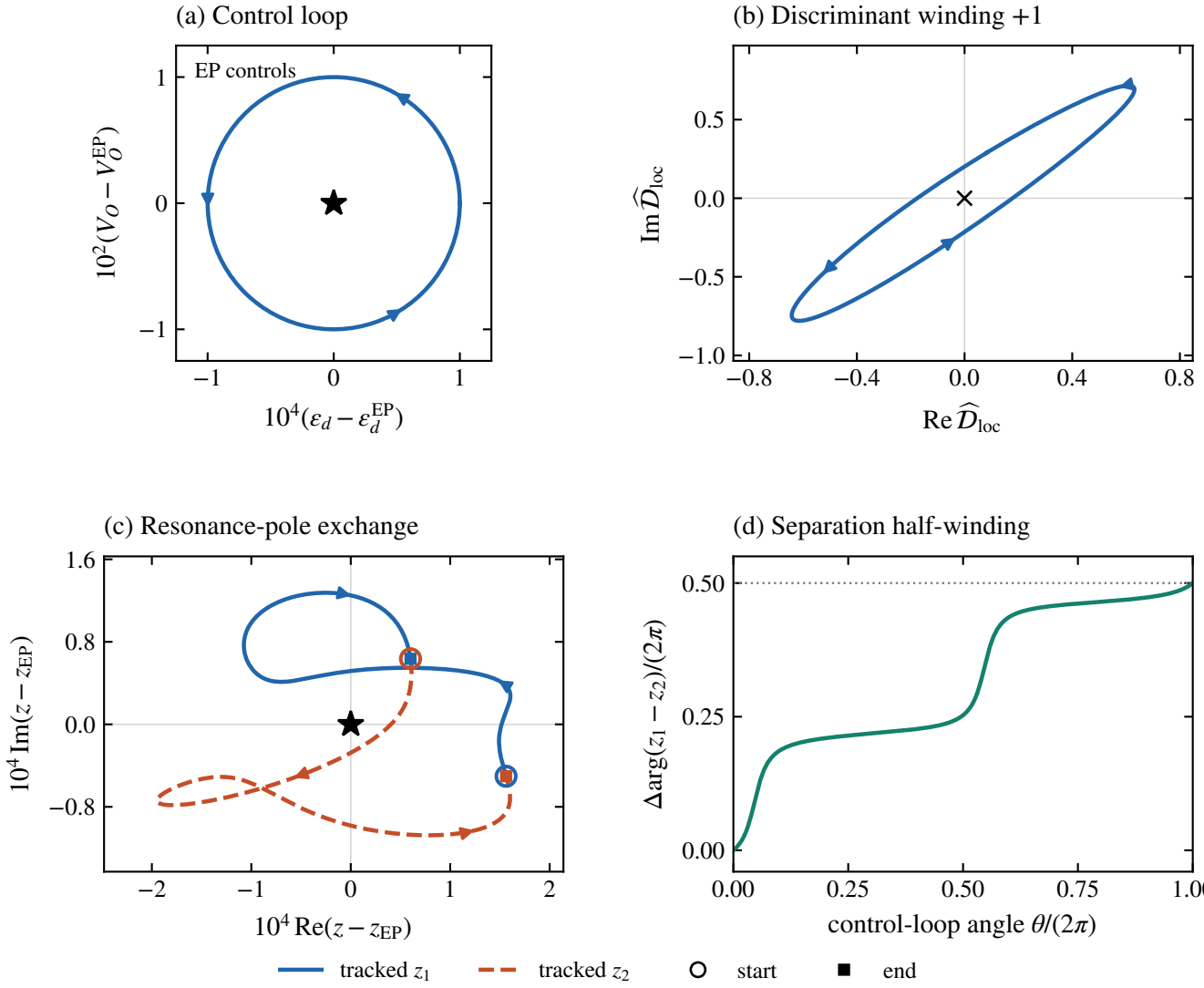
(a) Control loop
EP controls
$10^2(V_O - V_O^{\mathrm{EP}})$
$10^4(\varepsilon_d - \varepsilon_d^{\mathrm{EP}})$
(b) Discriminant winding +1
$\mathrm{Im}\,\widehat{D}_{\mathrm{loc}}$
$\mathrm{Re}\,\widehat{D}_{\mathrm{loc}}$
(c) Resonance-pole exchange
$10^4\,\mathrm{Im}(z - z_{\mathrm{EP}})$
$10^4\,\mathrm{Re}(z - z_{\mathrm{EP}})$
(d) Separation half-winding
$\Delta\arg(z_1 - z_2)/(2\pi)$
control-loop angle $\theta/(2\pi)$
tracked z_1
tracked z_2
start
end